

The cyclic sum fills every irreducible target representation

MathlibAnnex.CStarAlgebra.CAR.exists_surjective_selectedAtomicCyclicIsometry

theorem

Uses shell reconstruction and residual lines to turn an isometric CAR intertwiner into a unitary equivalence on the source.

Statement

Under the shell-family, generator and irreducibility assumptions below, there exist unit vectors $\zeta \in K$, $\eta_j \in K$ for $j \in I$, and a surjective complex-linear isometry $W : \mathcal{H} \rightarrow K$ such that

$$W(e_j \xi_j) = \eta_j, \quad V_j \eta_j = \zeta$$

for every j , and $W\pi(a) = \sigma(a)W$ for every $a \in A$. In particular, $\sigma = \rho \circ \iota$ is unitarily equivalent to the selected atomic source representation π .

Assumptions

Let A be the completed CAR algebra, the completion of the matrix stages $M_{2^n}(\mathbb{C})$ under $a \mapsto a \otimes I_2$. Let f_n be the image of the first diagonal matrix unit, so $f_0 = 1$ and f_n decreases through projections. The root state ω takes the value a_{00} on each matrix stage.

Choose one pure state φ_j from each unitary-equivalence class $j \in I$ of pure-state Gelfand-Naimark-Segal (GNS) representations, with root index o and $\varphi_o = \omega$. Write (H_j, π_j, ξ_j) for its complete complex GNS space, unital star representation and unit cyclic vector. Set $\mathcal{H} = \bigoplus_{j \in I} H_j$, $\pi = \bigoplus_j \pi_j$, and let $e_j : H_j \rightarrow \mathcal{H}$ be the coordinate embedding. No countability of I is assumed. Inner products are linear in the second argument.

A supplied shell family consists of unital complex star automorphisms α_j and elements $w_{j,n} \in A$. Put $f_{j,n} = \alpha_j(f_n)$. The identities are $\varphi_j \circ \alpha_j = \omega$, $w_{j,n}^* w_{j,n} = f_{j,n} - f_{j,n+1}$ and $w_{j,n} w_{j,n}^* = f_n - f_{n+1}$. At the root, α_o is the identity and $w_{o,n} = f_n - f_{n+1}$. The family is an input, not an existence conclusion.

Let L_j be unitaries on $\mathcal{H}$ satisfying $L_j \pi(f_{j,n} - f_{j,n+1}) = \pi(w_{j,n})$ for all j, n , and assume $L_o = I_{\mathcal{H}}$. Let $T = C^*(\pi(A), \{L_j\}) \subseteq B(\mathcal{H})$ be the norm-closed unital star algebra they generate. Write $\iota(a) = \pi(a)$ as an element of T , and $g_j \in T$ for the element represented by L_j . Let $\rho : T \rightarrow B(K)$ be a unital complex star representation on a complete complex Hilbert space, put $\sigma = \rho \circ \iota$, and put $V_j = \rho(g_j)$. Assume ρ is nonzero and irreducible: its only closed reducing subspaces are $\{0\}$ and K . This theorem does not assume the additional identity $L_j(e_j \xi_j) = e_o \xi_o$. The root normalization $L_o = I_{\mathcal{H}}$ is required.

Conclusion

The selected cyclic pieces exhaust the target Hilbert space. The source-restriction equivalence is the direct consequence obtained by viewing the surjective isometry as a unitary. The representation being assumed irreducible is ρ on T , not its generally reducible restriction σ on A .

The conclusion displayed here intertwines the CAR source. To obtain intertwining for every element of T , the full capture theorem also uses the prescribed action $L_j(e_j\xi_j) = e_o\xi_o$. No multiplicity assertion for arbitrary reducible target representations and no unconditional existence of a shell family is claimed.

Proof route

The common-root vector-state theorem supplies compatible unit vectors. Cyclic-sum assembly gives an isometry with closed source-reducing range M . Shell reconstruction splits each represented generator into a strong shell sum and a residual corner. Both parts and their adjoints preserve M . Norm-closed generation then makes M a nonzero reducing subspace for ρ , so irreducibility forces $M = K$.

▼ Proof steps

1. The common-root construction gives ζ and η_j of norm one with the selected CAR vector states, $V_j\eta_j = \zeta$, and fixedness under their respective flags. Assemble the cyclic pieces into W . Its range M is closed because $\mathcal{H}$ is complete and W is an isometry. The relations for a and a^* show that M reduces σ . Let P_j project onto the common range of $\sigma(f_{j,n})$ and let Q project onto the common range of $\sigma(f_n)$. Assembly gives $P_jM \subseteq \mathbb{C}\eta_j$.
2. Since $L_o = I_{\mathcal{H}}$, the represented root generator is $V_o = I_K$. Thus the root-index member η_o equals the common vector ζ . Consequently $\zeta \in M$ and $QM \subseteq \mathbb{C}\zeta$. Reconstruct $V_j = S_j + R_j$ with $R_j = V_jP_j$, $R_j^*R_j = P_j$, $R_jR_j^* = Q$ and $R_j = QR_jP_j$. The operators S_j and S_j^* are strong limits of finite sums of $\sigma(w_{j,n})$ and their adjoints. Closedness and source reduction make M invariant under both limits.
3. Fixedness and the vector transport give $R_j\eta_j = \zeta$ and $R_j^*\zeta = \eta_j$. For $y \in M$, the vector P_jy lies in $\mathbb{C}\eta_j$, so $R_jy \in \mathbb{C}\zeta \subseteq M$. For the adjoint, use $R_j^* = P_jR_j^*Q$ and $Qy \in \mathbb{C}\zeta$ to obtain $R_j^*y \in \mathbb{C}\eta_j \subseteq M$. Thus M reduces the residual corner and, together with the shell sum, every V_j .
4. The orthogonal projection onto M commutes with the represented source and every added generator. The commutation relation is stable under star-algebra operations and operator-norm limits, so M reduces all of $\rho(T)$. It contains the unit vector ζ and is therefore nonzero. Irreducibility gives $M = K$, which is exactly surjectivity of W . Its pointed-vector and source intertwining identities are retained.

Main citations

- Exact theorem and proof
- The supplied CAR shell-family data
- The atomic Hilbert sum and source representation
- A compatible family of selected vector states
- Isometric assembly and control of fixed projections
- Strong shell sums and residual corners
- Reduction extends through norm-closed generation
- The resulting unitary equivalence on the CAR source
- Full capture requires cyclic-vector transport on the model

- The root state with its purity proof

Lean source signature (exact)

```
theorem exists_surjective_selectedAtomicCyclicIsometry
  (family : RepresentativeShellFamily)
  (L : MathlibAnnex.CStarAlgebra.PureState.GNSClass Limit →
    MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert completedRootPureState
→L[ $\mathbb{C}$ ]
    MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert completedRootPureState)
  (hLunit : ∀ i, L i ∈ unitary
    (MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert completedRootPureState
→L[ $\mathbb{C}$ ]
    MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert completedRootPureState))
  (hLsource : ∀ i n, (L i).comp (selectedAtomicRepresentation
    (transportedFlag family i n - transportedFlag family i (n + 1))) =
    representedShellLink family i n)
  (hLroot : L completedRootPureState.classOf = 1)
  {K : Type v} [NormedAddCommGroup K] [InnerProductSpace  $\mathbb{C}$  K]
  [CompleteSpace K] (rho : Representation (AtomicTarget L) K)
  (hrho : rho.IsIrreducible) :
  ∃ (eta_o : K) (eta : MathlibAnnex.CStarAlgebra.PureState.GNSClass Limit → K)
    (W : MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert completedRootPureState
→ $_{i1}$ [ $\mathbb{C}$ ] K),
  Function.Surjective W ∧
  ||eta_o|| = 1 ∧
  (∀ i, ||eta i|| = 1) ∧
  (∀ i, W (MathlibAnnex.CStarAlgebra.PureState.selectedEmbedding completedRootPureState
i
    (MathlibAnnex.CStarAlgebra.PureState.selectedVector completedRootPureState i)) =
eta i) ∧
  (∀ i, (Unitary.linearIsometryEquiv
    (representedGeneratorUnitary L hLunit rho i) : K  $\simeq_{i1}$ [ $\mathbb{C}$ ] K)
    (eta i) = eta_o) ∧
  ∀ a,
  W.toContinuousLinearMap.comp (selectedAtomicRepresentation a) =
    ((restrictedRepresentation L rho) a).comp
    W.toContinuousLinearMap
```

In the source	Mathematical meaning
family : RepresentativeShellFamily	The given CAR shell family $(\alpha_j, w_{j,n})$, with $f_{j,n} = \alpha_j(f_n)$ and the initial and final support identities in the assumptions.
L; hLunit : ∀ i, L i ∈ unitary (...)	The given unitary family L_j on the selected atomic Hilbert space $\mathcal{H} = \bigoplus_j H_j$.

In the source	Mathematical meaning
<code>hLsource : ∀ i n, (L i).comp (...) = representedShellLink family i n</code>	For every j, n , the given shell relation is $L_j \pi(f_{j,n} - f_{j,n+1}) = \pi(w_{j,n})$, where $\pi = \bigoplus_j \pi_j$. Prescribed cyclic-vector transport is not assumed here; root normalization is the separate hypothesis <code>hLroot</code> below.
<code>rho : Representation (AtomicTarget L) K; hrho : rho.IsIrreducible</code>	The nonzero irreducible unital star representation $\rho : T \rightarrow B(K)$ of $T = C^*(\pi(A), \{L_j\})$, on a complete complex Hilbert space K in an arbitrary universe. Irreducibility means no proper nonzero closed reducing subspace.
<code>restrictedRepresentation L rho</code>	The source restriction $\sigma = \rho \circ \iota : A \rightarrow B(K)$, where $\iota(a) = \pi(a) \in T$. Nonzero irreducibility of ρ is not an assertion of irreducibility of this restriction.
<code>hLroot : L completedRootPureState.classOf = 1</code>	The root normalization $L_o = I_{\mathcal{H}}$ is an additional hypothesis. There is no hypothesis prescribing $L_j(e_j \xi_j) = e_o \xi_o$ in this declaration.
<code>∃ (eta_o : K) (eta : GNSClass Limit → K) (W : SelectedAtomicHilbert completedRootPureState →_i [C] K)</code>	One common vector ζ , family $(\eta_j)_j$, and complex-linear isometry $W : \mathcal{H} \rightarrow K$ satisfy all following clauses. The source name <code>eta_o</code> is ζ , not by definition the root-index member of <code>eta</code> .
<code>Function.Surjective W</code>	This same isometry fills all of K , so it can be viewed as a unitary without changing its map.
<code> eta_o = 1 ∧ (∀ i, eta i = 1)</code>	The common vector ζ and every η_j have norm one.
<code>∀ i, W (selectedEmbedding ... i (selectedVector ... i)) = eta i</code>	For each j , $W(e_j \xi_j) = \eta_j$ using the selected GNS coordinate embedding and unit cyclic vector.
<code>∀ i, (Unitary.linearIsometryEquiv (representedGeneratorUnitary L hLunit rho i) : K ≈_i [C] K) (eta i) = eta_o</code>	The same represented generator $V_j = \rho(g_j)$ sends η_j to the common ζ . The root normalization implies $V_o = I_K$ and hence $\eta_o = \zeta$ for the root-index family member.
<code>∀ a, W.toContinuousLinearMap.comp (selectedAtomicRepresentation a) = ((restrictedRepresentation L rho) a).comp W.toContinuousLinearMap</code>	For every $a \in A$, $W\pi(a) = \sigma(a)W$, where $\sigma = \rho \circ \iota$. This is intertwining of the CAR source; the statement does not yet intertwine all added target generators.

Further source notes: Here `Limit` is A , `completedRootPureState` is the root state ω bundled with its purity proof, and `SelectedAtomicHilbert ...` is $\mathcal{H}$. `hLsource` records the shell action; no hypothesis named `hLmap` occurs in this declaration.

▼ **Exact content identity**

Declaration: MathlibAnnex.CStarAlgebra.CAR.exists_surjective_selectedAtomicCyclicIsometry

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